CHIXUS MASTER SERVER ENGINE

System Blueprint: Leadership, Dedicated Inboxes & Server AI Architecture (v2.5.0)

FOUNDER: Chinaza Success Joseph | CO-FOUNDER: Ezenwa Christian Chimeremeze

Inboxes: chixus.system@gmail.com & chixus.support@gmail.com | Stack: Node/Express + FastAPI + SQLite

1. Executive Leadership & Dedicated System/Support Inboxes

- **Executive Founder:** Chinaza Success Joseph — Primary Architect, Lead Systems Engineer & Project Creator.
- **Co-Founder:** Ezenwa Christian Chimeremeze — Co-Architect, Hardware Systems & Operations Partner.
- **System Communications Inbox (chixus.system@gmail.com):** Designated official sender for all automated infrastructure traffic: 6-digit cryptographic verification codes, security alerts, access token issuances, and node ID/password recovery dispatches.
- **Support & Help Desk Inbox (chixus.support@gmail.com):** Designated support inbox for engineer inquiries, hardware fault escalation, contact requests, and issue ticketing.

2. Server-Side Gemini AI & Constitutional Safety Boundaries

- **Zero Client Exposure:** GEMINI_API_KEY is held exclusively on the backend server. The frontend client has zero API keys, making it 100% private and protected from scrapers or unauthorized browser extraction.
- **Render Cloud Deployment:** When deployed on Render, frontend requests (/api/ai/*) reach the Render backend service via standard HTTPS proxy, ensuring seamless communication while the backend holds the Gemini API.
- **Strict Rule 1: Zero Guessing & Anti-Hallucination:** CHIXUS AI must NEVER guess, extrapolate, or fabricate unverified facts, calculations, or metrics. If data is unknown, it delivers this friendly reliable message: "This inquiry falls outside verified telemetry baselines. For system integrity and safety, I do not speculate. Please refer to certified operational protocols or consult our engineering desk at chixus.support@gmail.com."
- **Strict Rule 2: Zero Architectural Disclosure:** CHIXUS AI must NEVER disclose internal frontend/backend source code, repository structure, secrets, or implementation details to users. Probing queries are politely deflected to leadership.
- **Strict Rule 3: Known Founders Recognition:** CHIXUS AI explicitly recognizes Founder: Chinaza Success Joseph and Co-founder: Ezenwa Christian Chimeremeze as its creators and official leaders.

3. Authentication, 6-Digit Code & Dual Recovery Protocol

- **Two-Factor 6-Digit Cryptographic Gateway:** Operators log in with Master Node ID + Password, triggering an automated 6-digit security verification code dispatched from chixus.system@gmail.com valid for 10 minutes (POST /api/auth/send-verification-code and POST /api/auth/verify-code).
- **Compulsory Dual Recovery Protocol (recover_credentials):** Verifies registered Phone Number and Email against permanent database records (with international and national prefix resilience). Returns BOTH original Master Node ID and account Password, and dispatches an official recovery email from chixus.system@gmail.com (POST /api/auth/recover-id and POST /api/send-recovery-email).
- **Zero Identity or Credential Loss:** Recovery never generates new accounts or alters permanent Master Node IDs. Operators can transfer recovered credentials straight into the terminal login form with 1 click.

4. Persistent Database Schema (SQLAlchemy & Master JSON)

A. Table: users — Permanent operator accounts & telemetry metrics

Column	Type	Constraints & Details
id	String	PRIMARY KEY, Indexed — Forever Master Node ID (e.g. CHI-4829-LAG)
username	String	Not Null, Indexed — Operator handle
full_name	String	Not Null — Legal full name

location	String	Not Null — Deployment sector / location
phone	String	Not Null, Indexed — Contact phone number
email	String	UNIQUE, Indexed, Not Null — Enforces 1 account per email
password_hash	String	Not Null — PBKDF2-HMAC-SHA256 salted password hash
headline	String	Default: "CHIXUS Hardware & Grid Operator"
avatar	String	Nullable — DiceBear identicon or custom avatar URL
inspections	Integer	Default: 0 (Auto-incremented on scan submission)
faults	Integer	Default: 0 (Auto-incremented when scan flags a fault)
accuracy	String	Default: "0.0%" (Recalculated automatically on scan log)
created_at	DateTime	Default: UTC timestamp

B. Table: posts — Shared multi-user network timeline feed

Column	Type	Constraints & Details
id	Integer	PRIMARY KEY, Auto-increment, Indexed
author_id	String	Not Null, Indexed — Operator Master Node ID
author_name	String	Not Null — Operator display name
content	Text	Not Null — Post text body
image_url	Text	Nullable — Image URL or Base64 Data URL
likes_count	Integer	Default: 0
comments_count	Integer	Default: 0
pinned	Boolean	Default: False (True for Administration Core post)
created_at	DateTime	Default: UTC timestamp

C. Table: scan_logs — Hardware & grid scan image ledger with 95% Gate

Column	Type	Constraints & Details
id	Integer	PRIMARY KEY, Auto-increment, Indexed
user_node_id	String	Not Null, Indexed — Operator Master Node ID
scan_title	String	Not Null — Inspection title
fault_detected	Boolean	Default: False
fault_type	String	Nullable — Fault classification
confidence_score	Float	Default: 95.0 — Gate verified at >= 95.0%
image_url	Text	Nullable — Visual telemetry snapshot URL
notes	Text	Nullable — Field notes
timestamp	DateTime	Default: UTC timestamp

D. Table: support_tickets — Help center tickets routed to chixus.support@gmail.com

Column	Type	Constraints & Details
id	String	PRIMARY KEY — Reference Ticket (e.g. TICK-4820-CHX)
operator_id	String	Nullable — Requester Node ID
name	String	Not Null — Submitter name
email	String	Not Null — Submitter email
category	String	Category (Engineering Inquiry, Hardware, Auth, Alert)
priority	String	Urgent (P1), High (P2), Normal (P3), Low (P4)
subject	String	Not Null — Inquiry subject
message	Text	Not Null — Detailed ticket body
status	String	open, investigating, resolved
assigned_inbox	String	Always assigned to chixus.support@gmail.com
created_at	DateTime	Default: UTC timestamp

5. Complete REST API Endpoint Reference

GET	/api/status	System health, inboxes, engine version, and database metrics.
POST	/api/auth/register	Register operator, enforce Unique Email Guard, bind Forever Node ID.
POST	/api/auth/login	Cross-device terminal authentication with Node ID & Password.
POST	/api/auth/send-verification-code	Dispatch 6-digit verification code from chixus.system@gmail.com.
POST	/api/auth/verify-code	Verify 6-digit cryptographic verification code and return auth token.
POST	/api/auth/recover-id	Dual recovery for BOTH Master Node ID & Password via registered phone & email.
POST	/api/send-recovery-email	Send credentials recovery email via chixus.system@gmail.com.
POST	/api/support/ticket	Open support ticket routed to chixus.support@gmail.com.
GET	/api/support/tickets	Retrieve list of logged support tickets.
POST	/api/notifications/dispatch-email	Dispatch security digest from chixus.system@gmail.com.
POST	/api/ai/scan-analyze	Server-side Gemini 3.8 Flash hardware fault scan analysis.
POST	/api/ai/diagnose	Server-side Gemini AI copilot with strict anti-hallucination rules.
GET	/api/users/search	Live Global Mesh directory search query.
POST	/api/posts	Publish post to shared network timeline.
GET	/api/posts	Fetch timeline posts (pinned core post first, then descending).
POST	/api/scans	Log scan telemetry with 95% Confidence Gate verification.
GET	/api/scans	Retrieve permanent scan ledger.
POST	/api/connections/request	Peer-to-peer operator mesh connection request.
GET	/api/connections	List incoming and outgoing connection requests.
PUT	/api/connections/:id	Accept or reject mesh connection request.
POST	/api/messages/send	Send persistent chat message to Chat Arena Vault.
GET	/api/messages/history	Retrieve chronological chat message history.

6. Render Deployment & Environment Variables Guide

- **Render Web Service Configuration:** Environment: Node.js (or Python 3.11). Build Command: "npm run build". Start Command: "npm start" (which executes tsx server.ts). Port: 3000.
- **Environment Variables on Render:** Set GEMINI_API_KEY=<your-secret-key> on the Render dashboard under Environment. The frontend makes zero external Gemini calls; it queries /api/ai/* on the Render backend service.
- **CORS Whitelist:** Allow origins includes https://chixus-system.netlify.app and local development ports. Credentials and headers are enabled.